

Rocket Agent Lab: classroom flight model

$$F_{\text{net}} = T - W - D = T - mg - \frac{1}{2}\rho v^2 C_D A$$

$$a = \frac{F_{\text{net}}}{m}$$

$$m_0 = m_{\text{dry}} + m_{\text{propellant}} + m_{\text{payload}} \quad m_f = m_{\text{dry}} + m_{\text{payload}}$$

$$\Delta v = I_{sp} g_0 \ln \left(\frac{m_0}{m_f} \right)$$

$$\Delta v_{\text{staged}} \approx G_{\text{stage}} I_{sp} g_0 \ln \left(\frac{m_0}{m_f} \right)$$

$$\frac{T}{W} = \frac{T}{m_0 g_0} > 1$$

$$q = \frac{1}{2}\rho v^2 \quad \rho(h) = \rho_0 e^{-h/H}$$

$$\text{payload margin} = m_{\text{payload,max}} - m_{\text{payload}}$$

$$\text{mission margin} = \Delta v_{\text{staged}} - \Delta v_{\text{required}}$$

The web app uses these equations as a transparent teaching model. It is not a full launch simulator: it is a hands-on layer for seeing how payload, fuel load, drag area, thrust-to-weight ratio, dynamic pressure, and mass ratio move together.